

Kees Dorst

Frame Innovation & Abductive Reasoning in Design

Professor of Design Innovation · University of Technology Sydney · TU Eindhoven



1 WHY THIS THINKER MATTERS

Dorst gives the academic backbone to reframing. His work on 'frame creation' and abductive reasoning explains why designers solve wicked problems differently from analysts — and how organisations can learn that move. He is the theory beneath Unit 6.

2 INTELLECTUAL FOUNDATION

Dutch design researcher, professor at UTS Sydney and TU Eindhoven. Drawing on Schön's reflective practice and design methodology, he formalised how designers create new problem frames to tackle open, complex ('wicked') challenges.

"The creation of frames is the key to design."

"Designers don't solve problems; they create new ways of seeing them."

3 FRAMEWORKS · TOOLS · MODELS

NAME	WHAT IT IS
Frame Innovation / Frame Creation	A nine-step method to create new problem frames
Abductive reasoning	Reasoning to what could be, given a desired outcome
Design as frame-making	Solving by proposing new ways of seeing
Wicked problems	Open, complex, contested challenges
Themes and paradox	Working from underlying themes to reframe paradoxes

4 HOW IT'S OPERATIONALIZED

- Analyse the paradox at the core of a stuck problem
- Search for deeper themes behind stakeholder positions
- Create and test new frames rather than optimise the old
- Use abductive 'what could be' reasoning to open options
- Apply frame creation to open, wicked challenges

5 LIMITATIONS & CRITIQUES

- Method is demanding and expertise-intensive
- Best suited to complex, not routine, problems
- Abstract for practitioners seeking quick tools
- Evidence largely case-based
- Requires time and senior buy-in

6 ENDURING IMPACT

- Frame Innovation a key academic text on reframing
- Formalised abduction in design reasoning
- Bridged design theory and organisational practice
- Influential in design-led innovation research
- Direct anchor for BCOBM126 Units 4 and 6 theory

SELECTED WORKS

FRAME INNOVATION <i>Create New Thinking by Design</i> DORST · 2015	THE DESIGN OF PROBLEMS <i>design methodology</i> DORST · 2010s	UNDERSTANDING DESIGN <i>175 Reflections</i> DORST · 2006	NOTES ON DESIGN <i>How Creative Practice Works</i> DORST · 2017
---	---	---	--

REFERENCES (HARVARD · CITE THEM RIGHT)

- Dorst, K. (2015) *Frame Innovation: Create New Thinking by Design*. Cambridge, MA: MIT Press.
- Dorst, K. (2011) 'The core of design thinking and its application', *Design Studies*, 32(6), pp. 521–532.